StockEvent Lens — Batch 018 Intelligence Dashboard

Articles 086–090 Analysis, Vector Embedding Sensitivity & Numerical Conflict Thresholds

Batch ID
`batch_18 (Articles 086-090)`

TEST 086
Original — 100% Base
NVIDIA $2B investment in Coherent (7.8M shares @ $256.80) & Merck holdings analysis.

TEST 087
Repeat — 97% Overlap
Re-syndicated account restating NVIDIA's $2B Coherent investment without new facts.

TEST 088
Update — +28% New Info
Finalizes long-term supply contracts & schedules initial optics delivery for next quarter.

TEST 089
Conflict — Terms Dispute
Contradicts equity terms: claims $3.5B investment, 12M shares purchased at $310.00.

TEST 090
Different — 0% Overlap
Automated IT sector whale options scanner alert starting standalone Group 036.

Technical Analysis: Dense Vector Blindness vs. Numerical Financial Contradiction Metrics

A critical flaw in automated financial news processing occurs when dense vector models (e.g., BAAI/bge-small-en-v1.5, OpenAI Ada-002) evaluate financial transaction terms. Because "NVIDIA", "Coherent", "investment", and "optical technology" dominate the semantic space, altering dollar amounts and share figures ($2.0B / 7.8M shares @ $256.80 vs. $3.5B / 12.0M shares @ $310.00 in test_089) still yields high cosine similarity (~0.92).

```
// Formula: Formal Conflict Detection Condition
Conflict_State = ( Cosine_Similarity(E_orig, E_cand) > τ_sim ) && ( ΔF_contradiction == 1.0 )
where τ_sim = 0.60, and ΔF_contradiction = 1.0 indicates a numerical transaction term dispute ($2.0B vs $3.5B equity value).
```

1. Vector Embedding Blindness
Dense embeddings cluster shared entity contexts closely. Consequently, "$2.0 billion at $256.80" and "$3.5 billion at $310.00" appear virtually identical to vector search engines, failing to catch critical financial discrepancies.

2. LLM Evaluation Override
To prevent corrupting financial metadata, the LLM evaluation layer conducts discrete entity-attribute extraction, forcing a hard conflict classification when $\Delta F_{contradiction} = 1.0$ is triggered.

Article Relationship & Percentage Breakdown (Batch 018)

FILE IDENTIFIER	PATTERN LABEL	KEYWORDS IDENTIFIED	SIMILARITY %	DELTA %	CLASSIFICATION RATIONALE
`test_086_original`	Original (%5=1)	NVIDIA Coherent stake, $2B investment, 7.8M shares, $256.80 price, Merck holdings	100%	Baseline	Establishes breaking story frame for NVIDIA's strategic Coherent investment and Merck portfolio analysis.
`test_087_repeat`	Repeat (%5=2)	Coherent NVIDIA investment, $2B funding, 7.8M shares, $256.80 per share	97%	0% New Facts	Restates exact investment details and share pricing from test_086 without introducing new state changes.
`test_088_update`	Update (%5=3)	**Formal volume supply contracts**, next-quarter production shipment schedule	82%	+28% Operational Facts	Advances event state by finalizing volume supply contracts and establishing an initial delivery schedule.
`test_089_conflict`	Conflict (%5=4)	**$3.5B investment dispute**, 12M shares, $310.00 per share claim	92% Match	$\Delta F_{contradiction}$ = 1.0	Directly disputes original transaction figures, asserting a $3.5B investment for 12M shares at $310.00.
`test_090_different`	Different (%5=0)	Benzinga IT sector whale options alert, NVDA calls, AAPL puts, options sweeps	0%	Group 036 Trigger	Zero domain overlap; automated daily options scanner report triggering standalone Group 036.

Group 035 — Consolidated Event Summary

4 Articles (086–089)

An institutional analysis evaluated strategic corporate investments by NVIDIA (Coherent, CoreWeave) and Merck (Alector, Entrada, Eikon) to identify key AI infrastructure and drug platform bottlenecks (Articles 086–087). Moving from equity positioning to operational execution, NVIDIA and Coherent finalized multiyear volume supply contracts and established an initial production shipment schedule for co-packaged optics starting next quarter (Article 088). However, key transaction terms remain in direct dispute: original reports confirmed NVIDIA invested $2.0 billion buying 7.8 million shares at $256.80 per share (Articles 086–087), whereas a competing account claims NVIDIA invested $3.5 billion to acquire 12.0 million shares at $310.00 per share (Article 089 ; $\Delta F_{contradiction}$ = 1.0).

✓ Update Preserved: Finalized Supply Contracts & Shipment Schedule
✓ Conflict Preserved: $2.0B / 7.8M @ $256.80 vs $3.5B / 12M @ $310.00
✓ Duplicate Removed: Article 087 Consolidated

Group 036 — Standalone Summary

1 Article (090)

Benzinga's options scanner detected high-volume whale activity across ten Information Technology stocks (Article 090). Notable trades included bullish call sweeps in NVDA ($225 strike) and MU ($1075 strike), alongside bearish put sweeps in AAPL ($335 strike) and bearish call sweeps in AMD ($630 strike).

✓ New Event Cluster Established ✓ Zero Entity Contamination from Group 035

Batch 018 Raw Data Source Inspector

JSON Format Verification & Evidence Code Snippets

`test_086_original.json` — Group 035

```
{
  "id": 61980736,
  "headline": "Alpha Buying: Follow the Money From NVI
  "symbols": ["ALEC", "COHR", "CRWV", "EIKN", "MRK", "
  "summary": "Coherent issued 7.8M shares to NVIDIA at
}
```

Establishes $2B / 7.8M shares @ $256.80 baseline.

`test_088_update.json` — Group 035

```
{
  "id": 990000088,
  "headline": "NVIDIA and Coherent execute long-term pu
  "symbols": ["COHR", "NVDA"],
  "summary": "NVIDIA and Coherent finalized formal mul
}
```

Adds volume supply contracts & delivery timetable.

`test_089_conflict.json` — Group 035

```
{
  "id": 990000089,
  "headline": "NVIDIA buys $3.5 billion stake in Coher
  "symbols": ["COHR", "NVDA"],
  "summary": "NVIDIA invested $3.5 billion into Cohere
}
```

Disputes equity valuation ($3.5B / 12M shares @ $310).